

Sotopia-RL on CSA: Deviations from the Published Method

Every place this implementation departs from Sotopia-RL (Yu, Qi et al., 2025), what forced it, and the flag that reverts it. Written so a reviewer can tell which results are the published method and which are ours.

reference arXiv:2508.03905 | target CSA, 99/9/42 scenarios | agents Qwen2.5-7B-Instruct | no API key required

1. What is unchanged

The method is Sotopia-RL. These are load-bearing and were not touched:

- Three stages: offline utterance-level reward design, then behaviour cloning, then a distilled reward model, then single-turn online GRPO.
- The reward form $r_t = G \cdot A(a_t, \tau)$ (Eq. 3): an episode-level score times a per-utterance attribution, per dimension, then normalised and averaged (Eq. 4).
- The attributor sees the FINISHED episode; the reward model sees only the prefix. That asymmetry is the whole point of distillation and is preserved exactly.
- The reward model is a scalar regressor trained with MSE (Eq. 5), not a preference model. The labels are attributed scalars, so no preference pairs exist or are needed.
- GRPO is single-turn: prompt is the dialogue history, completion is the next utterance, advantage is standardised within the group of G samples.
- The dialogue agent being trained is the one that speaks; there is no planner and no explicit reasoning trace, matching the paper's "utterance generation without explicit reasoning".

2. The changes, at a glance

Reason codes: **TASK** the CSA task demands it **BUDGET** the budget demands it **INTEGRITY** an integrity hole opens without it **GAIN** CSA offers something SOTOPIA cannot.

#	Vanilla Sotopia-RL	Here	Why	Revert with
1	GPT-4o attributor assigns $A(a_t, \tau)$	A computed from decisive_facts and reveal_turn	GAIN	n/a (structural)
2	Episode score G from an LLM judge	G from the executable verifier	GAIN	n/a
3	Rubric REL / KNO / GOAL	POOL / USE / COVER	TASK	--weights
4	COVER-analogue = social rubric dimension	advisor must be addressed AND then disclose	TASK	n/a
5	Eq.4 min-max, k unbound	normalised across the DATASET, so G survives	GAIN	--within_episode_norm
6	no integrity notion	leak or invalid schema zeroes the whole episode	INTEGRITY	gate flag
7	BC on GPT-4o self-play	BC on Qwen self-play, filtered by executable checks	BUDGET	--min_dca
8	reward model is required	optional: lookahead scores candidates exactly	GAIN	--reward_source rm
9	act/eliciting label unavailable	is_eliciting read from the utterance itself	TASK	n/a
10	full fine-tune, Qwen2.5-7B	LoRA r=16	BUDGET	n/a
10b	agent, policy and RM are separate models	ONE backbone; the three roles are adapters over it	BUDGET	n/a (structural)
11	SFT 500 epochs, RM 30 epochs	SFT 3, RM 4, early-stopped	BUDGET	--epochs
12	GRPO 16 generations	8 generations	BUDGET	--group 16

#	Vanilla Sotopia-RL	Here	Why	Revert with
13	RM selected on MSE	selected on within-episode pair ranking	TASK	n/a
14	two symmetric agents self-play	chair only; advisors frozen	INTEGRITY	n/a (see §5)
15	settling turn scored like any other	settlement scored EXACTLY by the verifier	GAIN	n/a

3. The reward is computed, not judged

Sotopia-RL's Eq. 3 multiplies an episode-level score by an LLM attributor's guess at how much each utterance contributed. Both factors are GPT-4o calls, both drift, and on SOTOPIA there is nothing to check either against. CSA states the answer: `decisive_facts` says which checks each private fact controls, and the transcript says which turn drew it out.

```
# Phi = every check some decisive fact controls; passed = those that passed

A_pool(t) = |U flips(f) : f ELICITED at turn t| / |Phi|
A_use(t)  = |{c in Phi : c passed, c in flips(f), f elicited at t}| / |Phi|
A_use(settle) += |{c in Phi : c passed, no elicited fact behind it}| / |Phi|
A_cover(t) = |advisors addressed at t that later disclosed| / |advisors|

r_{t,d}    = G_d . A_d(t) . gate(tau)      gate = 0 on a leak or invalid schema
```

- **Union, not sum.** The flips lists overlap, so summing lets one turn score above 1. Measured: summing produced a maximum of 1.222; the union gives exactly 1.0.
- **A_use only pays out if the settlement used the information.** If the settlement is wrong, no checks pass and every eliciting turn earns zero. The structure enforces that without a separate rule.
- **Volunteered facts earn nothing.** Credit requires the disclosure to follow an eliciting turn, which is the behaviour the benchmark exists to measure.
- **Provenance resolution is applied before scoring**, as in the other arms. Without it this arm runs a harsher provenance rule and the comparison breaks.

Measured over 297 real episodes (1,206 chair turns): 30.8%% of turns carry a non-zero label, mean 0.056, std 0.135, max 0.99 — dense enough to regress on, and every value reproducible without a single API call.

3.1 Eq. 4 normalisation: an ambiguity worth resolving

The paper writes $\min_k r_{k,d}$ without binding k . If k ranges over the turns of ONE EPISODE then G_d , constant across turns, cancels exactly:

$$(G.A_t - G.\min A) / (G.\max A - G.\min A) = (A_t - \min A) / (\max A - \min A)$$

and the episode score stops affecting the labels at all — a strong episode and a failed one produce identical targets. Ranging k over the dataset keeps G_d , which here is the verifier's own judgement and the most trustworthy signal available. Dataset-level is the default; `--within_episode_norm` reproduces the other reading for the ablation.

4. The rubric

SOTOPIA's seven social dimensions do not apply to a hidden-profile decision task, but CSA has its own, and two of the three map almost directly:

Paper	Here	What it measures	Non-zero
KNO (knowledge seeking)	POOL	did private information surface at all	13.4%
GOAL (goal completion)	USE	did it land in checks that passed	28.3%
REL (relationship)	COVER	were the holders actually drawn out	12.9%
—	aggregate r_t	equal-weight average, as the paper uses	30.8%

WHY COVER IS NOT "ADVISORS MENTIONED BY NAME"

The obvious definition — count advisors whose surname appears in a chair turn — saturates on turn 0, because the chair opens by greeting everyone. Measured that way it fired on 34.1% of turns and dominated the aggregate, rewarding politeness rather than elicitation. Worse, it was the densest of the three, so it was doing most of the work.

Here an advisor counts only once it has been addressed AND has subsequently disclosed something, crediting the turn that ASKED. Density drops to 12.9% and the aggregate from 51.7% to 30.8%, which is the correct trade: less signal, but signal that tracks the behaviour under test.

4.1 Eliciting-ness has to be read from the text

The planner baselines know whether a chair turn was an eliciting act because a planner chose the act. Here there is no planner, so `detectors_sr.is_eliciting` applies a lexical rule: the turn must address an advisor by name AND carry an interrogative. Requiring the name is what separates drawing information out of someone from thinking aloud.

This is a new source of measurement error, so it is measured rather than assumed. Against 650 turns carrying published act annotations: **accuracy 0.817, precision 0.845, recall 0.803**. `selftest.py` recomputes this on every run.

5. Self-play, and why it is chair-only

Sotopia-RL self-plays two symmetric agents. CSA is not symmetric: one chair holds no private facts and makes the decision, N advisors each hold one and do not.

THE NAIVE EXTENSION DESTROYS THE BENCHMARK

Training advisors on the same reward makes them state their private fact in the first utterance. Disclosure goes to 1, the hidden profile stops being hidden, the elicited fraction goes to 0 because nothing was elicited, and the chair needs no skill at all. The measured improvement would be real and entirely uninformative.

Advisors are therefore frozen and only the chair is trained. That is a genuine restriction of the published method and is reported as one.

6. The comparison caveat

THIS ARM IS NOT INTERCHANGEABLE WITH THE PLANNER BASELINES

Sotopia-RL fine-tunes the dialogue agent. The planner baselines hold the agent frozen and train a planner on top — precisely so the comparison isolates the planner. Putting all three in one table as though they answered the same question would be wrong.

Report two axes ("planner over a frozen agent" versus "fine-tuned agent, no planner"), or state plainly that the agent differs. Call counts and the frozen-advisor setup stay comparable; goal-completion numbers do not. Note also that GRPO's group sampling makes this arm several times more expensive per episode, so compare at matched budget rather than matched episode count.

7. Reproducibility

This package imports no code from the other baselines. It rebuilds the split from the raw scenarios and reimplements the verifier, then checks both against the published artefacts as data:

Check	Result
split reproduces the published 99/9/42, order included	exact
verifier reproduces cbar / pbar / disclosure on 297 episodes	exact, all 297
chair prompt free of private facts and oracle fields, 150 scenarios	clean
advisors see only their own private fact	clean
attribution stays within [0, 1]	max 0.9865
is_eliciting vs 650 annotated act labels	acc 0.817

Sotopia-RL: Yu, Qi, et al., *Sotopia-RL: Reward Design for Social Intelligence*, 2025 (arXiv:2508.03905). GRPO: Shao et al., 2024. Hidden profiles: Stasser & Titus, 1985. All figures in this document were recomputed from the CSA corpus by `selftest.py`; none are copied from an earlier summary.